TOGGLING IN $W_0(C)$ ROOT ORDER IS CONJUGATE TO ROWMOTION: RESOLUTION OF AIM CONJECTURE 3.2 AND THOMAS'S CONJECTURE

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Abstract. Toggling is a simple operation on a partially ordered set: it adds an element to a downward-closed subset, or removes one, whenever the result is still downward closed. Toggling every element in turn, from the top down, gives rowmotion, a map studied since the 1970s. Toggles also act on numerical labels, where they become rational maps, and there it is hard to tell whether a product of toggles ever returns to where it started. Even for rowmotion, this is known only for special posets.

In 2015, the problem list of the American Institute of Mathematics workshop on Dynamical Algebraic Combinatorics conjectured that one such product has order $2h$ on the root posets of types A, B, H_3 and $I_2(m)$, where h is the Coxeter number. This product toggles the positive roots in an order determined by a Coxeter element. We resolve the conjecture. We prove that the product is conjugate to rowmotion in every finite crystallographic type and in types H_3 (for an explicit labelling of its roots) and $I_2(m)$; this also proves a 2013 conjecture of H. Thomas. As a consequence, in every finite Weyl group the combinatorial version of the product has the same orbit structure as the Kreweras complement on noncrossing partitions. In the four types of the conjecture, it follows that the product returns to the identity after $2h$ steps. Its exact order is $2h$ in types A_n ($n \geq 2$) and $I_2(m)$ with m odd, but only h in types B_n, H_3 and $I_2(m)$ with m even. The only new computer calculations are in type H_3, and they are exact.

1. Introduction

1.1. **Toggles and rowmotion.** Let P be a finite poset. For $v \in P$, the *toggle* t_v acts on order ideals of P by adding v to an ideal I if $I \cup \{v\}$ is again an ideal, removing v if $I \setminus \{v\}$ is again an ideal, and doing nothing otherwise. Each toggle is an involution, and t_u and t_v commute unless u and v are joined by an edge of the Hasse diagram. The group they generate is the *toggle group* of P, introduced by Cameron and Fon-Der-Flaass [5] and developed systematically by Striker and Williams [27].

The most studied element of the toggle group is *rowmotion*: toggling every element once, from the top of the poset to the bottom. As a map on antichains it goes back to Brouwer and Schrijver [4], and Cameron and Fon-Der-Flaass [5] showed that it is this product of toggles. For the root poset Φ^+ of a finite Weyl group, Panyushev made three conjectures about rowmotion in its antichain form [21, Conj. 2.1]:

- it has order h if $w_0 = -1$ and order $2h$ otherwise, where h is the Coxeter number;
- its h-th power is the involution induced by $-w_0$;
- along every orbit, the average size of an antichain is $n/2$.

Armstrong, Stump and Thomas [2, Thm. 1.2] proved these conjectures by constructing a bijection, uniform in its definition, that carries rowmotion on antichains (nonnesting partitions) to the Kreweras complement on noncrossing partitions. They prove the first in the form $\mathrm{Row}^{2h} = \mathrm{id}$; together with an orbit of size h formed by the rank levels, observed by Panyushev, this gives the exact order.

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Einstein and Propp [11] lifted toggles to the *piecewise-linear* and *birational* settings, in which a toggle becomes a rational map on labellings of P (Section 2). Periodicity is much more delicate there. Grinberg and Roby [13, 14] proved that birational rowmotion is periodic on products of two chains and on several triangles. They also showed that it can have infinite order, already on small graded posets such as the Boolean lattice $[2]^3$ [13, §20], and they report that it has infinite order on the root poset of type D_4 [13, §21]. Musiker and Roby [19] gave a lattice-path formula for its iterates on products of two chains. Okada [20] proved periodicity on all minuscule posets, and Johnson and Liu [17] proved it on trapezoids, resolving a conjecture of Williams.

1.2. **Kreweras complements in the toggle group.** In his thesis [29, §4.4.1], Williams records an observation of H. Thomas. In type A_n, one of the toggle orders of Striker and Williams [27] is the order of the roots given by the c-sorting word of w_0. Thomas proposed a model of the Kreweras complement inside the toggle group of the root poset [29, Def. 4.4.4], as follows. Fix a Coxeter element c, and let $\beta_1, \ldots, \beta_N$ be the inversion sequence of the c-sorting word $w_0(c)$ of the longest element (Section 2). Toggling the elements of Φ^+ in this order defines Thomas's *nonnesting c-Kreweras complement*[1]

$$\mathrm{Krew}_c = \mathrm{Tog}_{\beta_1\beta_2\cdots\beta_N} = t_{\beta_N} \circ \cdots \circ t_{\beta_2} \circ t_{\beta_1},$$

so that β_1 is toggled first (Section 2). Thomas conjectured that these maps are all conjugate to rowmotion:

Conjecture 1.1 (Thomas; [29, Conj. 4.4.5])**.** *For every finite Weyl group W, any two of the maps* Krew_c *are conjugate in the toggle group of $\Phi^+(W)$, and they are conjugate to rowmotion.*

In type A, Dequêne, Frieden, Iraci, Schreier-Aigner, Thomas and Williams [7] constructed an explicit equivariant bijection between noncrossing partitions under the Kreweras complement and nonnesting partitions under the toggle product $\mathrm{Tog}_{\beta_1\cdots\beta_N}$. They call this product the Kroweras complement Krow_c [7, Def. 4.2], and it is exactly the map Krew_c above. In particular Krew_c has order $2h$ in type A [7, Cor. 1.6]. Their conjugation identity [7, Prop. 4.7], combined with the Striker–Williams conjugacy of promotion and rowmotion [27], yields Conjecture 1.1 in type A. They also treated type C by folding, at the level of equivariant bijections.

1.3. **The AIM problem.** The birational lift of Krew_c is the map K_c obtained by toggling the root poset in the same order, using birational toggles. The problem list [22] of the 2015 AIM workshop on Dynamical Algebraic Combinatorics records the following.

> Birational toggles in $w_0(c)$ root orders appear to continue to have order $2h$ in type A_n. This fails, for example, in type D_4 (as does birational rowmotion).
>
> **Conjecture 3.2.** *Birational toggles in $w_0(c)$ root orders have order $2h$ in the coincidental types A, B, H_3, $I_2(m)$.*

Two features make this harder than the corresponding statement for rowmotion.

First, an inversion order of w_0 is in general not a linear extension of the root poset, so K_c is not rowmotion, and the known periodicity results do not apply to it directly. The toggle order matters a great deal. On the four-element poset $[2] \times [2]$, birational rowmotion has order 4, but birational toggling in the order $(2,1),(2,2),(1,2),(1,1)$ gives a map of infinite order [13, Ex. 2.15], [11] (see Example 3.3).

Second, "has order $2h$" admits two readings: that $2h$ is the exact order, or that $2h$ is a period, i.e. $K_c^{2h} = \mathrm{id}$. The two readings already differ in rank two. The second is the form in which Armstrong, Stump and Thomas [2] state Panyushev's conjecture. It is also the form of

[1] This is a map on order ideals of Φ^+, not the Kreweras complement $w \mapsto cw^{-1}$ on noncrossing partitions used in [2].

Hopkins' conjecture [16, Conj. 4.40] that birational rowmotion on a coincidental root poset has order dividing $2h$. Panyushev's own formulation [21, Conj. 2.1] specifies the exact order: h if $w_0 = -1$, and $2h$ otherwise.

1.4. **Results.** Our first result proves Thomas's conjecture in every finite crystallographic type, for the combinatorial and the birational toggle groups alike, and extends it to the noncrystallographic coincidental types.

Theorem 1.2. *Let W be a finite Coxeter group of crystallographic type, or of type H_3 or $I_2(m)$ with Armstrong's root poset, labelled by roots as in Table 2 in type H_3. (In type $I_2(m)$ the labelling is irrelevant; see Section 7.) Let c be a Coxeter element. Then $\mathrm{Tog}_{\beta_1 \cdots \beta_N}$ is conjugate to rowmotion in the toggle group of $\Phi^+(W)$, for combinatorial, piecewise-linear and birational toggles. In particular Conjecture 1.1 holds.*

Combining Theorem 1.2 with what is known about the order of birational rowmotion settles the AIM problem.

Theorem 1.3. *Let W be of type A_n, B_n (with either the B_n or the C_n labelling of the root poset), H_3 (with the labelling of Table 2) or $I_2(m)$, and let c be any Coxeter element. Then $K_c^{2h} = \mathrm{id}$. The exact order of K_c is:*

- *$2h = 2(n+1)$ in type A_n for $n \geq 2$;*
- *$h = 2n$ in type B_n;*
- *$h = 10$ in type H_3;*
- *$\mathrm{lcm}(2, m)$ in type $I_2(m)$, that is, $2h$ for m odd and h for m even.*

Thus Conjecture 3.2 holds when "order $2h$" is read as a period. Read literally, it is false: already for $B_2 = I_2(4)$ the order is $4 = h$. Since $w_0 = -1$ exactly in types B_n, H_3 and $I_2(m)$ with m even, the orders in Theorem 1.3 follow Panyushev's dichotomy for combinatorial rowmotion.

Theorem 1.2 also explains the parenthetical remark in [22]. In type D_4 the map K_c is conjugate to birational rowmotion, so the two maps have the same order, and by [13, §21] that order is infinite.

For combinatorial toggles, Theorem 1.2 has a consequence that is close to the type A (and, by folding, type C) results of [7], and holds in every finite Weyl group.

Corollary 1.4. *Let W be an irreducible finite Weyl group and c a Coxeter element. Then Krew_c, acting on the order ideals of Φ^+ (which are in bijection with the nonnesting partitions of W), has the same orbit structure as rowmotion, and hence as the Kreweras complement on the noncrossing partitions $NC(W, c')$ for any Coxeter element c'. In particular, Krew_c has order h if $w_0 = -1$ and $2h$ otherwise. Moreover, there is an explicit bijection from $NC(W, c')$ to the order ideals of Φ^+ that carries the Kreweras complement to Krew_c.*

In types A and C, the equivariant bijections of [7] recalled above are direct. The bijection of Corollary 1.4 exists in every type but is less direct: it composes the uniform bijection of Armstrong, Stump and Thomas [2], whose well-definedness is proved case by case (by computer in the exceptional types), with the product of toggles that conjugates rowmotion to Krew_c. A direct bijection in every type would be desirable (Section 9).

1.5. **Method.** A toggle order induces an acyclic orientation of the Hasse diagram, and reversing all edges at a source (a *source-to-sink move*) amounts to conjugating by a single toggle. This principle goes back to the study of conjugacy of Coxeter elements [26, 12, 18], and it is stated for toggle groups in [10, §3] (see also [8]). By a theorem of Pretzel [23], two acyclic orientations

are related by source-to-sink moves if and only if they have the same circulation around every cycle.

So it suffices to show that the orientation induced by $w_0(c)$ has circulation zero around every cycle, as does the orientation induced by rowmotion (Section 3). Two classical facts about c-sorting words determine the orientation of every edge:

- Dyer's theory of reflection orders [9] reduces the orientation of a cover $\gamma \lessdot \gamma + \alpha_s$ to the relative position of γ and α_s;
- a compatibility property of Reading and Speyer [25] expresses that position through the sign of a skew-symmetric form Ω_c (Section 4).

Once the orientation of every edge is known, we check zero circulation on the small squares of the root poset, where opposite edges turn out to be parallel (Lemma 5.1), and show that these squares generate all cycles. In the simply laced types A, D and E this comes from a short local argument: two upper covers of a root always have a common upper cover (Lemmas 5.2 and 5.3). Types B_n and C_n are handled through a planar grid model of the root poset, and type F_4 is a short hand check using Lemma 5.3. Table 1 shows where each type is proved. Only the noncrystallographic type H_3 needs a computer. The orders are deduced in Section 8.

Type	Orientation of each edge	Circulation 0 on squares	Squares span all cycles
A_n, D_n, E_6, E_7, E_8	Lemmas 4.4, 4.5	Lemmas 5.1, 5.2	Corollary 5.4
B_n	Lemmas 4.4, 4.5	Lemma 5.1	Lemma 5.5
C_n	Lemma 4.4	Lemma 5.1	as in Lemma 5.5
F_4	Lemma 4.4	Lemma 5.1	Lemma 5.3
G_2, $I_2(m)$	the Hasse diagram is a tree, so there is nothing to check		
H_3	computer: circulation 0 on a cycle basis (Section 7)		

TABLE 1. Where the conjugacy $K_c \sim \mathrm{Row}$ is proved, type by type. Types C_n and F_4 are completed in the proof of Theorem 6.1, which also treats their squares with adjacent indices. In every type the conjugacy then follows from Corollary 3.4.

2. Preliminaries

2.1. **Root posets.** Let W be a finite Coxeter group with simple reflections $S = \{s_1, \dots, s_n\}$, simple roots $\Delta = \{\alpha_1, \dots, \alpha_n\}$, positive roots Φ^+, $N = |\Phi^+|$, and Coxeter number h. For crystallographic W, the *root poset* is Φ^+ ordered by $\beta \le \beta'$ if and only if $\beta' - \beta$ is a nonnegative integer combination of simple roots. Its covers are exactly the relations $\gamma \lessdot \gamma + \alpha_s$ with $\gamma, \gamma + \alpha_s \in \Phi^+$, and it is graded by height. For $I_2(m)$ and H_3 we use Armstrong's root posets [1], which Cuntz and Stump characterised as the unique posets with the expected enumerative properties [6]; they are described in Section 7. We write $\mathcal{H}(P)$ for the Hasse diagram of a poset P, viewed as an undirected graph.

2.2. **Coxeter elements and c-sorting words.** A *Coxeter element* is a product $c = s_{i_1} \cdots s_{i_n}$ of all simple reflections in some order; we fix such a reduced word. Consider the infinite word

$$c^\infty = s_{i_1} \cdots s_{i_n} s_{i_1} \cdots s_{i_n} \cdots .$$

Among its subwords that are reduced words for the longest element w_0, take the one whose sequence of positions is lexicographically first. This is the *c-sorting word* $w_0(c) = a_1 a_2 \cdots a_N$. Its inversion sequence

$$\beta_k = s_{a_1} s_{a_2} \cdots s_{a_{k-1}}(\alpha_{a_k}), \qquad 1 \le k \le N,$$

lists every positive root exactly once. We write $\beta \prec_c \beta'$ if β occurs before β' in this sequence.

Example 2.1. Let $W = A_3$ and $c = s_1s_2s_3$. Then $w_0(c) = s_1s_2s_3\,s_1s_2\,s_1$, and the inversion sequence is

$$\alpha_1,\ \alpha_1+\alpha_2,\ \alpha_1+\alpha_2+\alpha_3,\ \alpha_2,\ \alpha_2+\alpha_3,\ \alpha_3.$$

This is not a linear extension of the root poset: $\alpha_1+\alpha_2$ comes before α_2.

2.3. **Toggles.** Let $\widehat{P} = P \sqcup \{\hat{0}, \hat{1}\}$, with $\hat{0}$ below and $\hat{1}$ above every element. A *labelling* is a map $f : \widehat{P} \to \mathbb{K}$ to a field $\mathbb{K}$ (in practice a field of rational functions in the labels), with fixed boundary values $f(\hat{0}) = a$ and $f(\hat{1}) = b$. The *birational toggle* at $v \in P$ is

$$(T_v f)(v) = \frac{\sum_{u \lessdot v} f(u)}{f(v)\sum_{u \gtrdot v} 1/f(u)}, \qquad (T_v f)(x) = f(x) \quad (x \neq v),$$

where the sums run over the lower and upper covers of v in $\widehat{P}$ [11, 13]. The piecewise-linear toggles are its tropicalisation [11]. As in the combinatorial case, each T_v is an involution, and T_u and T_v commute unless u and v are adjacent in $\mathcal{H}(P)$ [13, §2]. These two facts are all we use about toggles in Sections 3–6, so those sections apply verbatim to combinatorial and piecewise-linear toggles.

For a total order $L = (x_1, \ldots, x_N)$ of P put

$$\mathrm{Tog}_L = T_{x_N} \circ \cdots \circ T_{x_2} \circ T_{x_1},$$

so that x_1 is toggled first. *Birational rowmotion* is $\mathrm{Row} = \mathrm{Tog}_L$ for L the reverse of a linear extension, so that higher elements are toggled first. We put

$$K_c = \mathrm{Tog}_{(\beta_1,\ldots,\beta_N)}.$$

Thus Krew_c, $\mathrm{Tog}_{\beta_1\cdots\beta_N}$ and K_c denote the same word in the toggles, read in the combinatorial, a general, and the birational toggle group, respectively.

3. Toggle orders, orientations and circulation

A total order L of P induces an acyclic orientation O_L of $\mathcal{H}(P)$: each edge points from the element toggled first to the element toggled later.

Lemma 3.1. (a) Tog_L *depends only on* O_L. *We write* $\mathrm{Tog}_O := \mathrm{Tog}_L$ *for any total order* L *with* $O_L = O$.

(b) *Let* x *be a source of an acyclic orientation* O, *and let* O' *be obtained from* O *by reversing all edges at* x, *so that* x *becomes a sink. Then* $\mathrm{Tog}_{O'} = T_x \circ \mathrm{Tog}_O \circ T_x$.

Proof. (a) The total orders inducing O_L are the linear extensions of the partial order generated by O_L. Any two linear extensions of a poset are connected by a sequence of swaps of adjacent, incomparable elements. Two elements that are incomparable in this partial order are not joined by an edge of $\mathcal{H}(P)$, so their toggles commute, and each swap leaves the product unchanged.

(b) Since x is a source, by (a) we may choose a total order $L = (x_1, \ldots, x_N)$ with $O_L = O$ and $x = x_1$. As T_x is an involution,

$$T_x \circ \mathrm{Tog}_L \circ T_x = T_x \circ T_{x_N} \circ \cdots \circ T_{x_2},$$

which is Tog of the order $(x_2, \ldots, x_N, x)$; this order induces O'. □

For a cycle C of $\mathcal{H}(P)$ with a chosen direction of traversal, the *circulation* $\nu_C(O)$ of an orientation O is the number of edges of C oriented along the direction of traversal minus the number oriented against it. A source-to-sink move does not change any circulation: a cycle through x uses two edges at x, one entering and one leaving x in the direction of traversal, and reversing both swaps their contributions.

Theorem 3.2 (Pretzel [23]; see also [8, Prop. 4.4])**.** *Two acyclic orientations of a connected graph are related by a sequence of source-to-sink moves if and only if they have the same circulation around every cycle.*

Example 3.3. On $[2]\times[2]$, the order $(2,1),(2,2),(1,2),(1,1)$ orients the unique 4-cycle so that, traversed as $(1,1)\to(1,2)\to(2,2)\to(2,1)\to(1,1)$, three edges point against the direction of traversal and one along it. Its circulation is -2, whereas rowmotion's is 0. So the two toggle products are not related by source-to-sink moves. Indeed, the first has infinite order and the second has order 4 [13, Ex. 2.15].

Call $g : P \to \mathbb{Z}$ a *potential* for an orientation O if $g(y) = g(x) + 1$ for every edge $x \to y$ of O. Summing $g(\text{head}) - g(\text{tail})$ around a cycle shows that an orientation with a potential has circulation 0 around every cycle. Conversely, on a connected graph, zero circulation around every cycle allows one to define a potential by integrating along paths. The following corollary needs only the easy direction of Theorem 3.2, and we include a short self-contained proof that also exhibits the conjugating element.

Corollary 3.4. *Let P be a graded poset with connected Hasse diagram. If O_L has circulation 0 around every cycle (equivalently, admits a potential), then* Tog_L *is conjugate to* Row *in the toggle group. More precisely, there are $x_1,\dots,x_k \in P$ with* $\mathrm{Tog}_L = T_{x_k}\cdots T_{x_1}\,\mathrm{Row}\,T_{x_1}\cdots T_{x_k}$.

Proof. Let g be a potential for O_L. In the orientation induced by rowmotion every edge points from the higher to the lower element, so $\varphi = -\mathrm{rank}$ is a potential for it. Along every edge both g and φ change by ±1, so $g-\varphi$ has constant parity on the connected graph $\mathcal{H}(P)$. Adding a constant to g, which does not change O_L, we may assume that $g \ge \varphi$ and $g \equiv \varphi \pmod 2$.

We modify φ step by step. At each stage φ is a potential for an acyclic orientation O_φ (with $x \to y$ when $\varphi(y) = \varphi(x)+1$), and Tog_{O_φ} is conjugate to Row by a product of toggles. Suppose $\varphi \neq g$. Let x minimise φ on the set $U = \{v : \varphi(v) < g(v)\}$, so that $g(x) \ge \varphi(x)+2$. If a neighbour y of x had $\varphi(y) = \varphi(x) - 1$, then $y \notin U$ by minimality, so $g(y) = \varphi(y) = \varphi(x) - 1 \le g(x) - 3$, contradicting $|g(x)-g(y)| = 1$. Hence every neighbour y has $\varphi(y) = \varphi(x)+1$, so x is a source of O_φ. Replacing $\varphi(x)$ by $\varphi(x)+2 \le g(x)$ turns x into a sink, and by Lemma 3.1(b) it conjugates Tog_{O_φ} by T_x. Each step decreases $\sum_v (g-\varphi)(v)$ by 2, so after finitely many steps $\varphi = g$, $O_\varphi = O_L$, and the claim follows. □

Since circulation is additive, it suffices to verify zero circulation on a set of cycles that spans the cycle space of $\mathcal{H}(P)$ over $\mathbb{Q}$.

Example 3.5. Figure 1 shows the orientation of Example 2.1. The numbers in brackets form a potential, so K_c is conjugate to rowmotion in type A_3 for this c.

4. Orientation of cover edges

In this section W is crystallographic. We determine the orientation of each edge $\gamma \lessdot \gamma + \alpha_s$ of $\mathcal{H}(\Phi^+)$ in two steps.

4.1. **Reduction to a comparison with a simple root.** The order $\prec_c$ is a *reflection order* in the sense of Dyer, since it is the inversion order of a reduced word for w_0.

Lemma 4.1 (Dyer [9, (2.2), (2.13)])**.** *If $\alpha \prec_c \gamma$ and $\beta = x\alpha + y\gamma \in \Phi^+$ with $x, y > 0$, then $\alpha \prec_c \beta \prec_c \gamma$.*

Corollary 4.2. *For a cover $\gamma \lessdot \gamma + \alpha_s$, the edge is oriented upward (that is, $\gamma \prec_c \gamma + \alpha_s$) if and only if $\gamma \prec_c \alpha_s$.*

Proof. The root $\gamma + \alpha_s$ is a positive combination of γ and α_s, so by Lemma 4.1 it lies between them in $\prec_c$. If $\gamma \prec_c \alpha_s$, then $\gamma \prec_c \gamma + \alpha_s$; if $\alpha_s \prec_c \gamma$, then $\gamma + \alpha_s \prec_c \gamma$. □

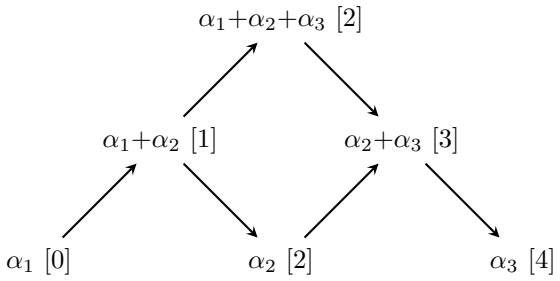


FIGURE 1. The root poset of A_3 with the orientation induced by $w_0(c)$ for $c = s_1s_2s_3$ (each edge points away from the root toggled first) and a potential in brackets.

4.2. **The form Ω_c.** Let $(\cdot,\cdot)$ be a W-invariant inner product, let a_{ij} be the Cartan matrix, and put $f_i = (\alpha_i,\alpha_i)/2$, so that $(\alpha_i,\alpha_j) = f_i a_{ij}$. Define a skew-symmetric bilinear form Ω_c on the root lattice by

$$\Omega_c(\alpha_i,\alpha_j) = \begin{cases} -(\alpha_i,\alpha_j) & \text{if } s_i \text{ precedes } s_j \text{ in } c,\\ \ \ (\alpha_i,\alpha_j) & \text{if } s_j \text{ precedes } s_i \text{ in } c,\\ 0 & \text{if } i=j \text{ or } a_{ij}=0. \end{cases}$$

It suffices to treat irreducible W. Then Ω_c is a positive scalar multiple of the form ω_c of Reading and Speyer [25], defined by $\omega_c(\alpha_i,\alpha_j) = -K(\alpha_i,\alpha_j)$ when s_i precedes s_j, where $K(\alpha_i,\alpha_j) = \delta(i)a_{ij}$. Here $\delta(i) > 0$ are the symmetrising constants of [25, §2.2], so that $\alpha_i^\vee = \delta(i)^{-1}\alpha_i$ and K is symmetric. Since both $(\delta(i)a_{ij})$ and $(f_i a_{ij})$ are symmetric, we have $\delta(i)/\delta(j) = f_i/f_j$ whenever $a_{ij} \neq 0$, so $\delta(i)/f_i$ is constant on an irreducible diagram. Note also that Ω_c depends only on c, not on the chosen reduced word.

Lemma 4.3 (Reading–Speyer [25, Prop. 3.11]; Reading [24, Cor. 4.4])**.** *If $\beta \prec_c \beta'$, then $\Omega_c(\beta,\beta') \geq 0$, with strict inequality unless the reflections s_β and $s_{\beta'}$ commute.*

Here Reading's theorem [24, Cor. 4.4] that w_0 is c-sortable is what makes Reading–Speyer's Proposition 3.11 applicable to the inversion order of $w_0(c)$.

Lemma 4.4. *Let $\gamma \lessdot \gamma+\alpha_s$ be a cover with $\Omega_c(\gamma,\alpha_s) \neq 0$. Then the edge is oriented upward if and only if $\Omega_c(\gamma,\alpha_s) > 0$.*

Proof. If the edge is oriented upward, then $\gamma \prec_c \alpha_s$ by Corollary 4.2, so $\Omega_c(\gamma,\alpha_s) \geq 0$ by Lemma 4.3. If it is oriented downward, then $\alpha_s \prec_c \gamma$, so $\Omega_c(\alpha_s,\gamma) \geq 0$, that is, $\Omega_c(\gamma,\alpha_s) \leq 0$. Since $\Omega_c(\gamma,\alpha_s) \neq 0$, its sign determines the orientation. □

Write $\gamma = \sum_t \gamma_t\alpha_t$ and $t \sim s$ for adjacency in the Dynkin diagram. Then

$$\Omega_c(\gamma,\alpha_s) = \sum_{t\sim s} \sigma_{ts}\,(\alpha_t,\alpha_s)\,\gamma_t, \qquad \sigma_{ts} = \begin{cases} -1 & \text{if } s_t \text{ precedes } s_s \text{ in } c,\\ +1 & \text{if } s_s \text{ precedes } s_t \text{ in } c. \end{cases} \tag{1}$$

In Example 2.1, with $(\alpha_i,\alpha_i) = 2$, we get $\Omega_c(\alpha_1,\alpha_2) = \Omega_c(\alpha_2,\alpha_3) = 1$. So the edge $\alpha_1 \lessdot \alpha_1+\alpha_2$ points up and the edge $\alpha_2 \lessdot \alpha_1+\alpha_2$ points down, as in Figure 1.

Whenever Lemma 4.4 applies to every cover, the orientation, and hence K_c, depends only on c and not on the reduced word chosen for it. In types G_2 and H_3 we checked all reduced words directly.

Lemma 4.5. *$\Omega_c(\gamma,\alpha_s) \neq 0$ for every cover $\gamma \lessdot \gamma+\alpha_s$ in types A_n, B_n, D_n and E_n.*

Proof. Simply laced types (normalise $(\alpha,\alpha)=2$ for all roots): since γ and $\gamma+\alpha_s$ are both roots of norm 2, we get $(\gamma,\alpha_s)=-1$. Expanding, $2\gamma_s-\sum_{t\sim s}\gamma_t=-1$, so $\sum_{t\sim s}\gamma_t$ is odd. By (1), $\Omega_c(\gamma,\alpha_s)=\sum_{t\sim s}\pm\gamma_t$ has the same parity, so it is nonzero.

Type B_n: take simple roots $\alpha_k=e_k-e_{k+1}$ $(k<n)$ and $\alpha_n=e_n$, with $(e_i,e_j)=\delta_{ij}$, so that $(\alpha_k,\alpha_{k\pm1})=-1$. The covers are

$$e_i-e_j\lessdot e_i-e_{j+1},\quad e_i-e_n\lessdot e_i,\quad e_i-e_j\lessdot e_{i-1}-e_j,\quad e_i\lessdot e_{i-1},\quad e_i\lessdot e_i+e_n,$$
$$e_i+e_j\lessdot e_{i-1}+e_j,\qquad e_i+e_j\lessdot e_i+e_{j-1}\ (j-1>i).$$

In all but the last family, exactly one neighbour t of s has $\gamma_t\neq0$, and $\gamma_t=1$, so $|\Omega_c(\gamma,\alpha_s)|=1$. In the last family, $s=j-1$ has the neighbours $j-2$ and j, with coefficients 1 and 2 in γ, so $|\Omega_c(\gamma,\alpha_s)|\in\{1,3\}$. □

5. Squares and the cycle space

A *square* of Φ^+ is a quadruple $(\gamma,\ \gamma+\alpha_s,\ \gamma+\alpha_t,\ \gamma+\alpha_s+\alpha_t)$ of positive roots with $s\neq t$. It is a 4-cycle in $\mathcal{H}(\Phi^+)$, consisting of two *s-edges* and two *t-edges*. We write it as $(\gamma;\ s,t)$.

Lemma 5.1. *Suppose that s and t are not adjacent in the Dynkin diagram, and that Ω_c is nonzero on all four edges of the square $(\gamma;\ s,t)$. Then opposite edges of the square are parallel, and its circulation is 0.*

Proof. Since $\Omega_c(\alpha_t,\alpha_s)=0$, we have $\Omega_c(\gamma+\alpha_t,\alpha_s)=\Omega_c(\gamma,\alpha_s)$. By Lemma 4.4 the two s-edges are therefore both oriented upward or both downward, and likewise the two t-edges. Going around the square, the two s-edges are traversed in opposite directions, and so are the two t-edges, so the circulation is 0. □

We now show that the squares generate all cycles. In simply laced types this follows from a local property of the root poset.

Lemma 5.2. *Let W be simply laced, and let $\gamma\in\Phi^+$ have two distinct upper covers $\gamma+\alpha_s$ and $\gamma+\alpha_t$. Then s and t are not adjacent in the Dynkin diagram, and $\gamma+\alpha_s+\alpha_t\in\Phi^+$.*

Proof. Normalise $(\alpha,\alpha)=2$ for all roots. As in the proof of Lemma 4.5, $(\gamma,\alpha_s)=(\gamma,\alpha_t)=-1$. Hence $(\gamma+\alpha_s,\alpha_t)=-1+(\alpha_s,\alpha_t)$, where (α_s,α_t) is -1 if s and t are adjacent and 0 otherwise. Two roots of norm 2 have inner product -2 only if they are negatives of each other, and $\gamma+\alpha_s\neq-\alpha_t$ because $\gamma+\alpha_s$ is positive. So s and t are not adjacent and $(\gamma+\alpha_s,\alpha_t)=-1$. The reflection in α_t therefore sends $\gamma+\alpha_s$ to $\gamma+\alpha_s+\alpha_t$, which is a positive root. □

In particular, no square of a simply laced root poset has adjacent s and t, since in a square $(\gamma;\ s,t)$ the root γ has the two upper covers $\gamma+\alpha_s$ and $\gamma+\alpha_t$.

The next lemma is a standard argument in the spirit of Newman's diamond lemma. For a saturated chain $C=(u_0\lessdot u_1\lessdot\cdots\lessdot u_k)$ of a poset P, write $[C]=\sum_i[u_{i-1},u_i]$ for the corresponding 1-chain of $\mathcal{H}(P)$, with each edge oriented upward.

Lemma 5.3. *Let P be a finite poset with a unique maximal element $\hat{m}$. Suppose that whenever an element u has two distinct upper covers v and v', some element w covers both v and v'. Then the cycle space of $\mathcal{H}(P)$ over $\mathbb{Z}$ is spanned by the 4-cycles $u-v-w-v'-u$ with $u\lessdot v\lessdot w$, $u\lessdot v'\lessdot w$ and $v\neq v'$.*

Proof. Let D be the span of these 4-cycles. Every element lies below $\hat{m}$, so saturated chains from any u to $\hat{m}$ exist. We first show, by downward induction on u, that $[C]-[C']\in D$ for any two saturated chains C,C' from u to $\hat{m}$. This is clear for $u=\hat{m}$. Otherwise let v and v' be the second elements of C and C'. If $v=v'$, apply the induction hypothesis at v. If $v\neq v'$, choose w covering v and v', and a saturated chain E from w to $\hat{m}$. By the induction hypothesis at v

and at v', $[C]$ and $[C']$ are congruent modulo D to the 1-chains of $u \lessdot v \lessdot w$ followed by E and of $u \lessdot v' \lessdot w$ followed by E. These differ by the 4-cycle on u, v, w, v', which lies in D.

Now fix, for each $u \neq \hat{m}$, an upper cover $p(u)$. The edges $\{u, p(u)\}$ form a spanning tree $\mathcal{T}$ of $\mathcal{H}(P)$: they are distinct, since u is the lower endpoint of $\{u, p(u)\}$, so there are $|P| - 1$ of them, and they connect every element to $\hat{m}$. Let C_u be the saturated chain $u \lessdot p(u) \lessdot p(p(u)) \lessdot \cdots \lessdot \hat{m}$. For an edge $u \lessdot v$ not in $\mathcal{T}$, its fundamental cycle with respect to $\mathcal{T}$ is $[u, v] + [C_v] - [C_u]$. This is the difference of the 1-chains of two saturated chains from u to $\hat{m}$, so it lies in D. The fundamental cycles form a basis of the cycle space. □

Corollary 5.4. *Let W be irreducible and simply laced. Then the squares span the cycle space of $\mathcal{H}(\Phi^+)$.*

Proof. The highest root is the unique maximal element of Φ^+ [3, Ch. VI, §1, no. 8, Prop. 25]. If γ has two distinct upper covers $\gamma + \alpha_s$ and $\gamma + \alpha_t$, then by Lemma 5.2 the root $\gamma + \alpha_s + \alpha_t$ covers both. So Lemma 5.3 applies. Its 4-cycles are exactly the squares: if w covers $\gamma + \alpha_s$ and $\gamma + \alpha_t$, then $w - \gamma$ is a sum of two simple roots and also lies in $\alpha_s + \mathbb{Z}_{\geq 0}\Delta$ and $\alpha_t + \mathbb{Z}_{\geq 0}\Delta$, so $w = \gamma + \alpha_s + \alpha_t$. □

In type B_n, the list of covers in the proof of Lemma 4.5 shows that every square has one of the forms

$$(e_i - e_j;\ i-1,\ j), \quad (e_i - e_n;\ i-1,\ n), \quad (e_i;\ i-1,\ n), \quad (e_i + e_j;\ i-1,\ j-1) \ \ (j-1 > i),$$

and in each, s and t are non-adjacent. For types B_n and C_n we show that the squares span the cycle space through a planar model, which we also use in Section 8.

Lemma 5.5. *In type B_n, the squares span the cycle space of $\mathcal{H}(\Phi^+)$.*

Proof. We embed Φ^+ in the lattice $\mathbb{Z}^2$ by $e_i - e_j \mapsto (i, j)$, $e_i \mapsto (i, n+1)$ and $e_i + e_j \mapsto (i, 2n+2-j)$. The image is $\Sigma_B = \{(i, p) : 1 \leq i \leq n,\ i+1 \leq p \leq 2n+1-i\}$, and each family of covers in the proof of Lemma 4.5 becomes a unit step inside Σ_B. Conversely, every unit step between two points of Σ_B arises this way. A step $(i, p) \to (i-1, p)$ adds α_{i-1}. A step $(i, p) \to (i, p+1)$ adds a simple root that depends only on p: it is α_p for $p < n$, α_n for $p \in \{n, n+1\}$, and α_{2n+1-p} for $p \geq n+2$.

So $\mathcal{H}(\Phi^+)$ is identified with the *induced* grid graph on Σ_B, whose edges are all unit steps between points of Σ_B. A unit square of the lattice has all four corners in Σ_B if and only if it is the image of a square of Φ^+. The region Σ_B is connected and both row-convex and column-convex.

Consider the plane drawing of this graph. We claim that every bounded face is a unit square. Suppose that a bounded face F contains two adjacent open unit cells. The unit segment between them is not an edge of the graph, so at least one of its endpoints q is not in Σ_B. This lattice point lies on no edge, so it belongs to F. Since F is bounded, each horizontal ray from q leaves F, so it meets the boundary of F. That boundary consists of edges between points of Σ_B, and it meets the horizontal line through q only at points of Σ_B. So Σ_B contains points on both sides of q in its row, and row-convexity forces $q \in \Sigma_B$, a contradiction. Hence every bounded face is a single unit cell, whose boundary is the image of a square of Φ^+. For a connected plane graph, the boundaries of the bounded faces span the cycle space, which proves the lemma. □

Proposition 5.6. *In types A_n, B_n, D_n and E_n, K_c is conjugate to* Row *for every Coxeter element c.*

Proof. It suffices to treat irreducible W. By Lemma 4.5, Ω_c is nonzero on every cover. In types A_n, D_n and E_n, no square has adjacent indices (Lemma 5.2), and the squares span the cycle space (Corollary 5.4). In type B_n, the same holds by the list of squares above and Lemma 5.5.

By Lemma 5.1 every square has circulation 0, hence so does every cycle, and Corollary 3.4 applies. □

6. All crystallographic types

Theorem 6.1. *For every finite crystallographic Coxeter group W and every Coxeter element c, the map $\mathrm{Tog}_{\beta_1 \cdots \beta_N}$ is conjugate to rowmotion in the combinatorial, piecewise-linear and birational toggle groups of $\Phi^+(W)$. In particular Conjecture 1.1 holds.*

Proof. It suffices to treat irreducible W. Types A_n, B_n, D_n and E_n are Proposition 5.6.

Type C_n. Take $\alpha_k = e_k - e_{k+1}$ $(k < n)$ and $\alpha_n = 2e_n$, with $(e_i, e_j) = \delta_{ij}$, so that $(\alpha_k, \alpha_{k+1}) = -1$ for $k \leq n-2$ and $(\alpha_{n-1}, \alpha_n) = -2$. The covers are

$$e_i - e_j \lessdot e_i - e_{j+1}, \quad e_i - e_n \lessdot e_i + e_n, \quad e_i - e_j \lessdot e_{i-1} - e_j, \quad e_i + e_j \lessdot e_{i-1} + e_j,$$

$$e_i + e_j \lessdot e_i + e_{j-1} \ (j - 1 > i), \qquad e_i + e_{i+1} \lessdot 2e_i, \qquad 2e_i \lessdot e_{i-1} + e_i.$$

The map $e_i - e_j \mapsto (i, j)$, $e_i + e_j \mapsto (i, 2n + 1 - j)$, $2e_i \mapsto (i, 2n + 1 - i)$ identifies $\mathcal{H}(\Phi^+(C_n))$ with the induced grid graph on the region Σ_B of Lemma 5.5, so the squares span the cycle space by the same argument.

On each cover above, the nonzero contributions $|(\alpha_u, \alpha_s)|\gamma_u$ $(u \sim s)$ in (1) form one of the multisets $\{1\}$, $\{2\}$ or $\{1, 2\}$. Hence $\Omega_c(\gamma, \alpha_s) \in \{\pm 1, \pm 2, \pm 3\}$ is nonzero, and Lemma 4.4 applies.

Squares with non-adjacent s, t are handled by Lemma 5.1. The squares with adjacent s, t are exactly $(e_k + e_{k+1};\ k - 1,\ k)$.

- For the two s-edges, $\Omega_c(\gamma, \alpha_s)$ and $\Omega_c(\gamma + \alpha_t, \alpha_s)$ both come from the single neighbour k, with coefficients 1 and 2, so they have the same sign.
- For the two t-edges, $|\Omega_c(\gamma, \alpha_t)| = 2$ and $|\Omega_c(\alpha_s, \alpha_t)| = 1$, so adding α_s to γ does not change the sign.

Hence these squares also have parallel opposite edges and circulation 0. So every cycle has circulation 0, and Corollary 3.4 applies.

Type G_2. With α_1 short, the positive roots are α_1, α_2, $\alpha_1 + \alpha_2$, $2\alpha_1 + \alpha_2$, $3\alpha_1 + \alpha_2$ and $3\alpha_1 + 2\alpha_2$. Both simple roots are covered by $\alpha_1 + \alpha_2$, and the four non-simple roots form the chain $\alpha_1 + \alpha_2 \lessdot 2\alpha_1 + \alpha_2 \lessdot 3\alpha_1 + \alpha_2 \lessdot 3\alpha_1 + 2\alpha_2$, so the Hasse diagram is a tree. It has no cycles, so Corollary 3.4 applies to every toggle order.

Type F_4. Number the simple roots as in [3, Plate VIII], so that α_1, α_2 are long and α_3, α_4 are short, and normalise $(\alpha_1, \alpha_1) = 2$ and $(\alpha_3, \alpha_3) = 1$. Then $(\alpha_1, \alpha_2) = (\alpha_2, \alpha_3) = -1$ and $(\alpha_3, \alpha_4) = -\frac{1}{2}$. We write roots by their coordinates, so $1221 = \alpha_1 + 2\alpha_2 + 2\alpha_3 + \alpha_4$. The 24 positive roots are listed in [3, Plate VIII], and everything below can be checked by hand from that list.

First, Ω_c is nonzero on every cover $\gamma \lessdot \gamma + \alpha_s$. Some neighbour t of s has $\gamma_t \neq 0$: the support of the root $\gamma + \alpha_s$ is connected and contains s, so if no neighbour of s were in the support of γ, then γ, and hence $\gamma + \alpha_s$, would be a multiple of α_s. For $s = 1$ or $s = 4$ there is only one neighbour, so (1) has a single nonzero term. For $s = 2$: since α_2 is long, α_2-strings have length at most 2, so $\gamma - \alpha_2 \notin \Phi$ and $(\gamma, \alpha_2) = \langle \gamma, \alpha_2^\vee \rangle = -1$, that is, $\gamma_1 + \gamma_3 = 2\gamma_2 + 1$; so $\gamma_1 \neq \gamma_3$, and the terms $\pm\gamma_1$ and $\pm\gamma_3$ cannot cancel. For $s = 3$, the terms are $\pm\gamma_2$ and $\pm\frac{1}{2}\gamma_4$, and the ten covers $\gamma \lessdot \gamma + \alpha_3$, with $\gamma = 0001, 0100, 0110, 1100, 0111, 1110, 1111, 1221, 1222, 1232$, all have $\gamma_4 \neq 2\gamma_2$. So Lemma 4.4 applies.

Second, the squares span the cycle space. The roots with more than one upper cover are 0010, 0100, 0110, 0111, 0120, 1110, 0121, 1120, 1121 and 1221, and in each case $\gamma + \alpha_s + \alpha_t$ is a root for every two upper covers $\gamma + \alpha_s$, $\gamma + \alpha_t$. The highest root 2342 is the unique maximal element, so Lemma 5.3 applies, and as in Corollary 5.4 its 4-cycles are the squares.

Third, every square has circulation 0. Squares with non-adjacent indices are handled by Lemma 5.1. The squares with adjacent indices are $(\gamma; 3, 4)$ for $\gamma = 0110$, 1110 and 1221. The two 4-edges get their Ω_c-values only from the neighbour 3, with coefficients γ_3 and $\gamma_3 + 1$, so they have the same sign. For the two 3-edges, $|\Omega_c(\gamma, \alpha_3)| \geq 1$ (it is 1 for 0110 and 1110, and $\frac{3}{2}$ or $\frac{5}{2}$ for 1221), while $|\Omega_c(\alpha_4, \alpha_3)| = \frac{1}{2}$, so adding α_4 to γ does not change the sign. As in type C_n, opposite edges are parallel, every cycle has circulation 0, and Corollary 3.4 applies. □

Remark 6.2 (Independent computational checks)**.** The following computations are not used in the proofs above; they confirm them independently. For every ordering of S in types E_6, E_7, E_8, F_4 and G_2 (720, 5040, 40320, 24 and 2 orderings), the orientation induced by $w_0(c)$ admits a potential, and the same holds in types A_n ($n \leq 9$) and B_n, C_n, D_n ($n \leq 8$). In types E, F and G the square cycles have full rank modulo the prime $2^{31} - 1$, hence over $\mathbb{Q}$, and every square has parallel opposite edges. In type F_4, Ω_c is nonzero on every cover for all 24 orderings. The hypothesis of Lemma 5.3 holds in every crystallographic type we tested, including the non-simply-laced ones.

Proof of Corollary 1.4. By Theorem 6.1 and Corollary 3.4 (which applies verbatim to combinatorial toggles), $\mathrm{Krew}_c = \pi \circ \mathrm{Row} \circ \pi^{-1}$, where $\pi = t_{x_k} \cdots t_{x_1}$ is a product of toggles computed from a potential of the $w_0(c)$ orientation. So π is a bijection on order ideals that carries Row to Krew_c, and the two maps have the same orbit structure. The map I sending an antichain to the order ideal it generates carries rowmotion on antichains, which is Panyushev's map Pan, to Row [5]. For a bipartite Coxeter element c', the Main Theorem of Armstrong, Stump and Thomas [2, §1.3] gives an explicit bijection α from antichains to $NC(W, c')$ with $\alpha \circ \mathsf{Pan} = \mathsf{Krew} \circ \alpha$, where $\mathsf{Krew}(w) = c'w^{-1}$. Then $\pi \circ I \circ \alpha^{-1}$ carries Krew to Krew_c. Conjugation $w \mapsto xwx^{-1}$ carries $NC(W, c')$ with its Kreweras complement to $NC(W, xc'x^{-1})$, and all Coxeter elements are conjugate, so any Coxeter element may be used in place of c'.

For the order: Armstrong, Stump and Thomas [2, Thm. 1.2] proved that $\mathsf{Pan}^{2h} = \mathrm{id}$ and that Pan^h is the involution induced by $-w_0$, which is the identity exactly when $w_0 = -1$. Panyushev observed that the rank levels of Φ^+, together with the empty antichain, form an orbit of size h (see [2, §1.3]). So the order of Pan, and hence of Krew_c, is h if $w_0 = -1$ and $2h$ otherwise.

7. The noncrystallographic types

7.1. **Type $I_2(m)$.** Armstrong's root poset of $I_2(m)$ is $\{x, y\} \oplus (z_1 < \cdots < z_{m-2})$: two incomparable minimal elements below a chain. Its Hasse diagram is a tree, so it has no cycles. By Corollary 3.4, every toggle order of it is conjugate to Row. In particular, K_c is conjugate to Row whatever labelling of the poset by roots of $I_2(m)$ is used. It remains to find the order of Row.

Write f' for the labelling after the toggles performed so far. Rowmotion toggles $z_{m-2}, \ldots, z_1$, and then x and y. When x and y are toggled, they receive the new labels $D/(Uf(x))$ and $D/(Uf(y))$, where $D = f(\hat{0})$ and $U = 1/f'(z_1)$ are the same for both. Hence $r = f(x)/f(y)$ is mapped to $1/r$. Put $\Sigma_{xy} = f(x) + f(y)$ and $\Sigma^*_{xy} = 1/f(x) + 1/f(y)$. Then the new values are $(D/U)\,\Sigma^*_{xy}$ and $(U/D)\,\Sigma_{xy}$, so $k = \Sigma_{xy}\Sigma^*_{xy} = (1 + r)^2/r$ is invariant, and Σ_{xy} is mapped to $kD/(U\Sigma_{xy})$.

Comparing with the toggle rule shows that $(\Sigma_{xy}, z_1, \ldots, z_{m-2})$ evolves exactly as birational rowmotion on a chain of $m - 1$ elements whose bottom label is kD. That map has exact order m: a chain of $m - 1$ elements is $(m - 1)$-graded, so this follows from [13, Prop. 7.3]. Since $f(x)$ and $f(y)$ are recovered from r and Σ_{xy}, and $r \mapsto 1/r$ has period 2, the order of Row on $I_2(m)$ is $\mathrm{lcm}(2, m)$.

7.2. **Type H_3.** We use the simple roots of H_3 with $m(s_1,s_2)=5$ and $m(s_2,s_3)=3$, and put $\tau=(1+\sqrt{5})/2$. In [6, §3.2, Fig. 2] the 30 positive roots of D_6, written in an orthonormal basis $v_1,\dots,v_6$, are grouped into 15 pairs, one for each positive root of H_3. From each pair take the vector that is lexicographically greater in the coordinates $v_1,\dots,v_6$, and order these 15 roots by the root order of D_6. The result is Armstrong's H_3 root poset: it has $32=\mathrm{Cat}(H_3)$ order ideals and rank sizes $3,2,2,2,2,1,1,1,1$.

Each element is labelled by the H_3 root it represents. For a D_6 root $a=(a_1,\dots,a_6)$ in simple-root coordinates, the vector $(a_5+\tau a_3,\ a_2+\tau a_4,\ a_1+\tau a_6)$ is 1 or τ times a unique positive root of H_3 in simple-root coordinates, and that root is the label of a. Table 2 lists the result. As Williams notes, for H_3 the labelling is part of the data of the problem. In this poset covers are not in general of the form $\beta \lessdot \beta+\alpha_s$ (for example $(1,0,0)\lessdot(\tau,1,0)$), so Sections 4 and 5 do not apply, and we argue by computer.

D_6 root		H_3 root	D_6 root		H_3 root	D_6 root		H_3 root
100000	$\mapsto$	$(0,0,1)$	011000	$\mapsto$	$(\tau,1,0)$	111101	$\mapsto$	$(1,\tau,\tau)$
010000	$\mapsto$	$(0,1,0)$	111000	$\mapsto$	$(\tau,1,1)$	111111	$\mapsto$	(τ,τ,τ)
110000	$\mapsto$	$(0,1,1)$	011100	$\mapsto$	$(1,\tau,0)$	111211	$\mapsto$	$(\tau,\tau+1,\tau)$
001000	$\mapsto$	$(1,0,0)$	011110	$\mapsto$	$(\tau,\tau,0)$	112211	$\mapsto$	$(\tau+1,\tau+1,\tau)$
			111100	$\mapsto$	$(\tau,\tau+1,1)$	122211	$\mapsto$	$(\tau+1,2\tau,\tau)$
			111110	$\mapsto$	$(\tau+1,\tau+1,1)$			

TABLE 2. The H_3 root poset inside $\Phi^+(D_6)$ (D_6 roots in simple-root coordinates) and its labelling by positive roots of H_3 (simple-root coordinates).

Conjugacy. There are six orderings of S, giving four Coxeter elements up to commutation. For each, the orientation induced by $w_0(c)$ has circulation 0 around every cycle of a cycle basis, so K_c is conjugate to Row by Corollary 3.4.

Periodicity. Grinberg and Roby report a Sage computation showing $\mathrm{Row}^{10}=\mathrm{id}$ on this poset [13, §21]. We verified this deterministically. Let $x=(x_1,\dots,x_{15})$ be the generic labelling, with boundary labels 1. We computed $F=\mathrm{Row}^5(x)$ and $G=\mathrm{Row}^{-5}(x)$ exactly, as reduced fractions in $\mathbb{Z}[x_1,\dots,x_{15}]$, using multivariate arithmetic with gcd normalisation; the largest polynomial has 11247 terms. The result is $F=G$ identically, so $\mathrm{Row}^{10}=\mathrm{id}$ as rational maps. General boundary labels reduce to $(1,1)$ by the rescalings $f\mapsto\lambda f$ and $f(v)\mapsto t^{\mathrm{rank}(v)}f(v)$, which commute with all toggles. The same computation gives $\mathrm{Row}^5\neq\mathrm{id}$, and $\mathrm{Row}^2\neq\mathrm{id}$ at an explicit rational point, so the order is exactly $10=h$. An independent randomised identity test modulo four large primes agrees.

8. Orders of rowmotion and the proof of Theorem 1.3

By Theorem 6.1 and Section 7, K_c is conjugate to Row, so the two maps have the same order. It remains to recall the order of birational rowmotion on each root poset.

Type A_n. The root poset $\Phi^+(A_n)$ is the triangle $\Delta(n+1)$ of Grinberg and Roby [14, Rem. 18.3], on which Row has order exactly $2(n+1)=2h$ for $n\geq 2$ [14, Cor. 18.9].

Type B_n. The root poset $\Phi^+(B_n)$ is isomorphic to the triangle $\mathrm{NEtri}(2n)$ [14, Rem. 19.3], for which Grinberg and Roby conjectured that Row has order $2n$ [14, Conj. 19.5]. This follows from the work of Johnson and Liu [17, Cor. 5.3] on the trapezoid

$$T_{r,s}=\{(I,J)\in[r+s-1]\times[s] : I-J<r,\ I+J>s\}$$

(with the product order), on which they show that rowmotion has order exactly $r+s$. The map $(i,p)\mapsto(p-1,n+1-i)$ is an isomorphism from the grid model of $\Phi^+(B_n)$ in Lemma 5.5 onto $T_{n,n}$, since it sends up-steps to unit steps. Johnson and Liu state their result for positive real

labellings with boundary labels 1. For graded posets, general boundary labels reduce to this case by renormalisation [11], and since $\mathbb{R}^N_{>0}$ is Zariski dense, $\mathrm{Row}^{2n} = \mathrm{id}$ holds as an identity of rational maps. So the order on $\Phi^+(B_n)$ is exactly $2n = h$. The type C_n labelling gives an isomorphic poset.

Type $I_2(m)$. The order is $\mathrm{lcm}(2, m)$ by Section 7. This also follows from [13, Prop. 7.3], as noted in [13, §21].

Type H_3. The order is $10 = h$ by Section 7.

In each coincidental type the order divides $2h$. This proves $K_c^{2h} = \mathrm{id}$ and Theorem 1.3. As a sanity check, floating-point iteration of birational rowmotion at a random positive labelling recovers these orders for A_n ($n \le 5$), B_n ($n \le 6$) and $I_2(m)$ ($3 \le m \le 14$).

Remark 8.1 (Hopkins' conjecture)**.** Hopkins conjectured that birational rowmotion on every root poset of coincidental type has order dividing $2h$ [16, Conj. 4.40]. He noted that Grinberg and Roby had established it for A_n, H_3 and $I_2(m)$ (for H_3 by a computer calculation [13, §21]), leaving type B_n open. As recently as 2023 Grinberg and Roby still described periodicity on all coincidental root posets as open [15]. The results recalled above settle it. The B_n case is due to Johnson and Liu [17], via the isomorphism $\Phi^+(B_n) \cong T_{n,n}$. Our contribution is a deterministic exact certificate in type H_3.

9. Concluding remarks

Remark 9.1 (Related work)**.** The conjugacy of products of involutions whose orders differ by source-to-sink moves is classical for Coxeter elements [26, 12, 18], and for toggle groups it appears in [10, §3], where toggle actions on noncrossing partitions and their homomesies are also studied. For minuscule posets, Okada [20] proved that birational Coxeter-motion is periodic and conjugate to rowmotion. Thomas and Williams [28] developed rowmotion as a sequence of local moves ("slow motion") on trim lattices. What is new here is a uniform description of the toric-equivalence class, in the sense of [8], of the $w_0(c)$ orientation, through the sign of Ω_c.

Remark 9.2 (Open questions)**.** (1) Our proof is computational only in type H_3. A conceptual treatment of Armstrong's noncrystallographic root posets would be desirable. (2) In type H_3, K_c depends on the chosen labelling of the root poset by roots. It would be interesting to know which labellings give a map conjugate to rowmotion. (3) Corollary 1.4 gives an equivariant bijection between noncrossing and nonnesting partitions in every type, but an indirect one. A direct construction in every type, extending the bijections of [7] in types A and C, would be desirable.

Code

All computations are in the folder `anc/` that accompanies this paper (Python and C++). It is also available at `https://doi.org/10.5281/zenodo.23602944` and `https://github.com/samsonmaximus/toggle-w0c-rowmotion`. The proofs use computation only in type H_3 (Section 7). The other computations, described in Remark 6.2 and Sections 7 and 8, are independent checks. The file `anc/README.md` lists, for each computational claim in the paper, the file, the command and the expected output. The files cover:

- the root posets, c-sorting words and orientations (`roots.py`, `orient.py`);
- the exact potential and square checks (`fastcheck.cpp`, `squares.cpp`, `omega.cpp`);
- the check of the hypothesis of Lemma 5.3 and of the span of the squares (`diamond.py`);
- the H_3 conjugacy check (`h3run.py`);
- the H_3 certificate (`h3_sym_half.py`, with its run log);
- the floating-point sanity checks (`roworder.py`, `i2m.py`).

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Use of AI

The research presented here is the result of human–AI interaction with Claude Opus 5.5 (Anthropic). I chose the problem and directed the project. The AI system found the approach through Pretzel's theorem, developed the proofs and wrote the code. Comments from an expert reader on an earlier version led to the hand proofs in Sections 5 and 6 that remove every computer calculation from the crystallographic case. Separate AI sessions refereed each proof, checked each citation against its source and recomputed every computational claim with independent code. The problems were not routine: Conjecture 3.2 had been open since 2015, and Thomas's conjecture since 2013, with its type A case settled only recently [7]. I take full responsibility for the contents and welcome detailed questions about the role of AI in this work.

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